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# Study of diversity of freshwater mollusca in the Artificial Reservoir, Getalsud located in Ranchi Jharkhand

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**Abstract-** This study has been conducted in Getalsud Dam, Ranchi, Jharkhand to explore the moulluscan diversity in the artificially created dam. This study has been done season wise during March 2022 to March 2023. Sampling has been done season wise by dividing the study period into three season (pre- monsoon, monsoon and post - monsoon). During the study, a total of 12 species, which belongs to 7 families were collected. During the study the species collected showed a variation both in the quantity as well as in quality. Viviparidae and Lymnaeidae showed greatest abundance in the three season justifying the eury nature of their existence. This study was done so that molluscan diversity can be explored more and more.

**Key words: Mollusca, season, Getalsud dam, Viviparidae, Lymnaeidae**

## INTRODUCTION

Mollusca is the second largest animal group in the world. These are found in both water and land, showing an amphibious nature. Mollusca are found in both marine and freshwater. Mollusca are the second most successful group in the world's biodiversity after insects.[1,2] Only 279 families out of 586 are found in the Indian subcontinent and out of 3600 species about 2300 are found in marine water and the rest in fresh water.[3] Mollusca are found in every ecosystem of the world from the Arctic to Antarctica, depths of the ocean, ponds, lakes, rivers and mountain ranges.[4] Mollusca live in lotic and lentic water. They act as environmental and biological indicators. They help in aquatic food chain to maintain the aquatic ecosystems. The body is soft but strong and covered with calcium carbonate.They can be easily identified by their shell character. Mollusca is an important source of food for other animals like fishes,birds and mammals including human beings.

Getalsud Dam is an Artificial Reservoir present in Ormanjhi Ranchi Jharkhand India. It was built on the Subernarekha River in 1971. It is geographically located in the north eastern edge of Chotanagpur Plateau. It fulfills the drinking water demand of the people of Ranchi city and also water from this dam is used for industrial requirements. Fish farming is also done in this dam. Nearly 500 families are living in the villages near this dam and are dependent on this dam for their livelihood. A wide variety of Macrobenthic population has been reported in this water body, molluscans apparently showing the maximum density among macrobenthic population in all seasons. So this study has been carried to analyses the diversity showed by the macrobenthic population of mollusca.

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## MATERIALS & METHODS

### Study area

Getalsud Dam comes under the Ranchi district, Jharkhand situated between 23°27'24''N 85°33'19''E and 23°45'66''N 85°55'52''E. The Getalsud Dam is a manmade reservoir that came into existence in 1971. It was constructed across the river Subernarekha. It is a popular picnic spot for the residents of Ranchi and Ramgarh district. It is situated in Rukka block because of which it is also locally known as Rukka Dam.

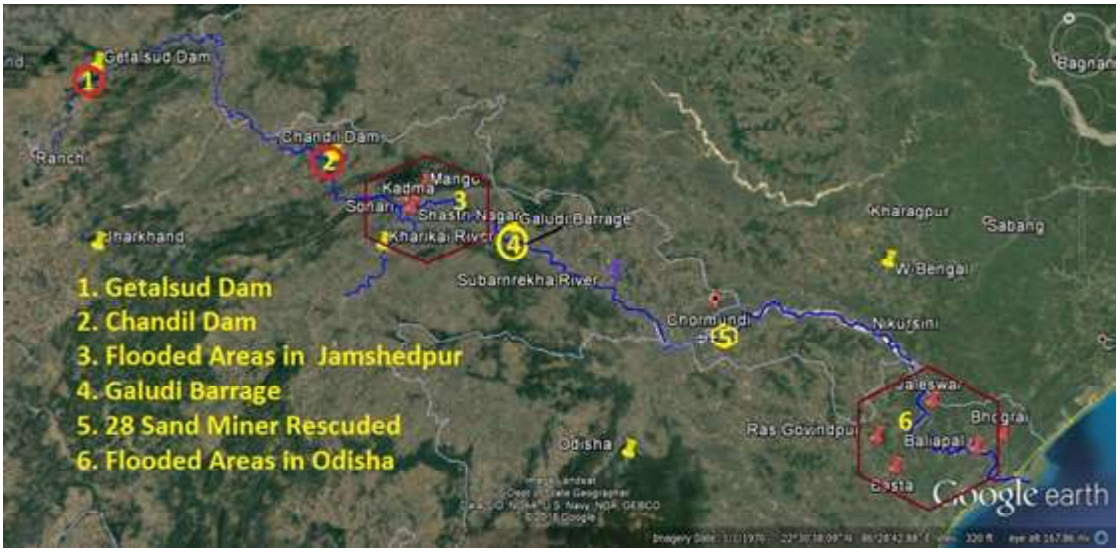



**Fig. 1- Satelite view of water bodies near study area (Source-Google earth)**

The present study was conducted from March 2022 to March 2023 seasonwise (pre-monsoon, monsoon and post-monsoon)

Four sampling sites were chosen for mollusca collection:-

(1) Near village area
(2) Near pool side area
(3) Near forest area
(4) Near fish farming area



**Fig. 2- Site 1 Near village area**



**Fig. 3- Site 2 Near pool side area**



**Fig. 4- Site 3 Near forest area**




**Fig. 5- Site 4 Near fish farming area**

## COLLECTED SAMPLES









**Collection, preservation and identification**

- Samples were collected from every site through hand pick up and net.
- After this the samples were washed in fresh water to remove mud and other substances.
- After washing the samples was preserved in 70% alcohol.
- The species has been identified from the book, Indian freshwater mollusca by Mitra *et al.,* (2005)[5], and some species was identified by scientists of ZSI (Zoological Survey of India)

## RESULT & DISCUSSION

**Table 1 - List of molluscs from Getalsud Reservoir Ranchi, Jharkhand**

| Sl. No. | Class | Order | Family | Species |
|---|---|---|---|---|
| 1 | Gastropoda | Hygrophila | Lymnaeidae | *Lymnaea (Pseudosuccinea) chlamys* |
| | | | | *Lymnaea (Pseudosuccinea) acuminata* (Lamarck, 1822) |
| | | | Planorbidae | *Indoplanorbis exustus* (Deshayes,1834) |
| | | | | *Gyraulus convexiusculus* (Hutton, 1949) |
| | | Architaenioglassa | Viviparidae | *Bellamya Bengalensis typica* (Lamarck, 1822) |
| | | | | *Bellamya bengalensis f. annandalei* (Kobelt, 1909) |
| | | | Ampullariidae | *Pila globosa* (Swainson, 1822) |
| | | Littorinimorpha | Bithyniidae | *Gabbia stenothyroides* (Dohrn, 1857) |
| | | | | *Gabbia orcula* (Frauenfeld,1862) |
| | | Sorbeoconcha | Thiaridae | *Melanoides tuberculata* (Müller, 1774) |
| 2 | Bivalvia | Unionoida | Unionidae | *Lamellidens marginalis* (Lamarck,1819) |
| | | | | *Lamellidens corrianus* (Lea, 1834) |

**Table 2- Sitewise details of collected species**

| Taxa | Site 1 | Site 2 | Site 3 | Site 4 |
|---|---|---|---|---|
| **Class - Gastropoda** | | | | |
| **Family- Lymnaeidae** | | | | |
| *Lymnaea (Pseudosuccinea) chlamys* | + | - | + | + |
| *Lymnaea (Pseudosuccinea) acuminata* | - | - | + | + |
| **Family - Planorbidae** | | | | |
| *Indoplanorbis exustus* | - | - | + | + |
| *Gyraulus convexiusculus* | - | - | + | - |
| **Family-Viviparidae** | | | | |
| *Bellamya bengalensis typica* | + | + | + | + |
| *Bellamya bengalensis f. annandalei* | + | + | + | + |
| **Family -Ampullariidae** | | | | |
| *Pila globosa* | + | - | + | - |
| **Family- Bithyniidae** | | | | |
| *Gabbia stenothyroides* | - | + | + | + |
| *Gabbia orcula* | + | + | + | + |
| **Family- Thiaridae** | | | | |
| *Melanoides tuberculata* | + | - | + | + |
| **Class- Bivalvia** | | | | |
| **Family - Unionidae** | | | | |
| *Lamellidens marginalis* | - | + | + | + |
| *Lamellidens corrianus* | - | + | + | - |

12 species of gastropod and bivalves have been recorded from 4 sampling stations of Getalsud Dam. Mollusca are known as the most diverse and dominant benthic fauna in both lentic and lotic region. 2 classes of mollusca gastropods and bivalves have been found in Getalsud Dam. Researchers observed 18 gastropods and 7 bivalves's species from north Bihar region India. Sudarshan R. Markad and Akhila S. Pillai (2021)[6] from Ujani wetland Maharashtra, India have studied 11 species of gastropod and 9 species of bivalvia. Species of Mollusca have been observed by Angsuman Chanda's survey from paschim medinipur district of West Bengal.[7] From Maharashtra Jadhav Amit and Zod Priyanka have recorded 13 taxas of mollusca from Tawajar river.[8] 12 species of gastropod and bivalves have been recorded from 4 sampling sites at Getalsud dam which includes 10 species of gastropod and 2 species of bivalves. The first sample was collected from near the village. Here dam water is used to irrigate paddy fields. Due to its proximity to paddy fields, *Pila globosa*, Swainson (1828)[9] has also been recorded here and species of Viviparidae class have also been found in large numbers in this area. The second stations was closer to the pool where anthropogenic activity was higher. Boating and other water activities are done in this area during pre-monsoon period. The amount of sand soil in this area are more. On this site the species of unionidae, Viviparidae and bithyniidae were found in every less quantity. The third sampling site was near the forest. In this site, species of Lymnaeidae, Viviparidae class have been recorded. In large numbers along with species of Planorbidae, Ampilaridae, Tharidae, Unionidae family have also been recorded. Animals and birds are present in large numbers in this area, which shows that they are also dependent on the grass and aquatic fauna for food and water.

The fourth sample was collected from the area near Fish farming, here birds are present in large numbers which are waiting to eat the numerous fish and aquatic animals. In this area the species of Lymnaeidae, Planorbidae, Viviparidae, Ampullaridae, Bithyniidae, Thiaridae and Unionidae families were recorded.

Different types of soil are found on different sites of the dam. Some area had more sandy soil and some had more textured soil. But mollusca was dominating everywhere. Due to Getalsud dam being a very large reservoir, fish diversity, aquatic animals, aquatic insects, aquatic weeds, mollusca and benthic fauna are available in quite large numbers.

## CONCLUSION

During the study in Getalsud dam 12 species of mollusca were recorded which belongs to 7 families, 2 classes of mollusca gastropods and bivalves show greater abundance in Getalsud Dam. This is quite a big reservoir, many people depend on this reservoir for their livelihood and drinking water. The ecological health of water bodies can be assessed by studying mollusca. this species can be utilized as a bioindicators for water pollution and aquatic ecosystems health by studying mollusca, we can contribute to better management and conservation of mollusca in this area which wil be helpful for further studies.

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